

◆ SECTION 5: Electric Dipole & Dipole Moment (Q51–60)

51. The SI unit of dipole moment is:
- A) C·m
 - B) C/m
 - C) N·m
 - D) J/C
52. The torque on a dipole in uniform electric field is maximum when:
- A) $\theta = 0^\circ$
 - B) $\theta = 90^\circ$
 - C) $\theta = 180^\circ$
 - D) $\theta = 45^\circ$
53. Dipole moment direction is from:
- A) Positive to negative
 - B) Negative to positive
 - C) Along field lines
 - D) Perpendicular to field
54. Electric field along the axial line of a dipole is:
- A) Half of equatorial field
 - B) Twice the equatorial field
 - C) Equal to equatorial field
 - D) Zero
55. Which of the following is an example of permanent dipole?
- A) HCl molecule
 - B) Hydrogen atom
 - C) Na⁺ ion
 - D) Electron
56. Torque on a dipole is given by:
- A) $p \cdot E \cdot \cos\theta$
 - B) $p \cdot E \cdot \sin\theta$
 - C) p/E
 - D) $p/E \cdot \theta$
57. Dipole in non-uniform field experiences:
- A) Torque only
 - B) Force only
 - C) Both force and torque
 - D) None
58. For an ideal dipole, charge separation is:
- A) Very large
 - B) Zero
 - C) Very small
 - D) Infinite
59. Electric field on equatorial line is:
- A) Along dipole axis
 - B) Perpendicular to axis
 - C) Zero
 - D) Maximum

60. Dipole in uniform field has potential energy:
- A) $U = -p \cdot E \cdot \cos\theta$
 - B) $U = p \cdot E \cdot \sin\theta$
 - C) $U = p \cdot E \cdot \theta$
 - D) $U = 0$

◆ **SECTION 6: Electric Potential & Potential Difference (Q61–75)**

61. Electric potential is:
- A) Force per unit charge
 - B) Work done per unit charge
 - C) Energy per unit mass
 - D) Charge per unit distance
62. SI unit of electric potential:
- A) Volt
 - B) Ampere
 - C) Coulomb
 - D) Newton
63. Potential difference between two points A and B is:
- A) W/q
 - B) q/W
 - C) F/q
 - D) q/F
64. Work done in moving a charge along equipotential surface:
- A) Maximum
 - B) Zero
 - C) Minimum
 - D) Depends on charge
65. Potential at infinity due to a point charge is:
- A) Zero
 - B) Maximum
 - C) Minimum
 - D) q/r
66. Potential is a:
- A) Vector
 - B) Scalar
 - C) Tensor
 - D) None
67. Electric potential at distance r from point charge q :
- A) $V = kq/r^2$
 - B) $V = kq/r$
 - C) $V = kq \cdot r$
 - D) $V = kq \cdot r^2$
68. If potential difference is doubled, energy of a charge:
- A) Doubles
 - B) Halves

- C) Quadruples
D) Remains same
69. Relation between electric field E and potential V :
A) $E = -dV/dr$
B) $E = dV/dr$
C) $E = V \cdot r$
D) $E = V/r^2$
70. Potential due to dipole at equatorial line:
A) Zero
B) Maximum
C) Half of axial
D) Equal to axial
71. Moving a charge along equipotential surface requires:
A) Maximum work
B) Zero work
C) Half work
D) Depends on path
72. Electric potential inside a conductor:
A) Varies linearly
B) Zero
C) Constant
D) Maximum
73. Relation between electric field and potential difference in uniform field:
A) $E = V/d$
B) $E = d/V$
C) $E = V \cdot d$
D) $E = d \cdot V^2$
74. Unit of potential difference is:
A) N/C
B) Volt
C) Joule
D) Ampere
75. Potential energy of a charge q at potential V :
A) $U = qV$
B) $U = V/q$
C) $U = q/V$
D) $U = V^2/q$

◆ SECTION 7: Capacitance & Combination of Capacitors (Q76–95)

76. Capacitance is defined as:
A) Q/V
B) V/Q
C) $Q \cdot V$
D) V/Q^2
77. SI unit of capacitance:
A) Farad

- B) Coulomb
 - C) Volt
 - D) Ohm
78. Capacitance of parallel plate capacitor depends on:
- A) Area of plates & separation
 - B) Material only
 - C) Charge only
 - D) Voltage only
79. For capacitors in series, equivalent capacitance is:
- A) Sum of capacitances
 - B) Reciprocal of sum of reciprocals
 - C) Product of capacitances
 - D) None
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- A) Sum of capacitances
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 - D) None
81. Energy stored in a capacitor:
- A) $\frac{1}{2} CV^2$
 - B) CV
 - C) $\frac{1}{2} C^2V$
 - D) CV^2
82. Dielectric increases capacitance because:
- A) Reduces potential difference
 - B) Increases charge storage
 - C) Reduces field
 - D) All of above
83. Dielectric constant of vacuum:
- A) 1
 - B) 0
 - C) >1
 - D) <1
84. Capacitance of spherical capacitor depends on:
- A) Radii of shells
 - B) Charge
 - C) Potential
 - D) Both B & C
85. Capacitance of parallel plate capacitor is directly proportional to:
- A) Plate separation
 - B) Area of plates
 - C) Square of voltage
 - D) $1/\text{Area}$
86. Capacitance of parallel plate capacitor is inversely proportional to:
- A) Plate separation
 - B) Plate area
 - C) Dielectric constant
 - D) Voltage

87. Combination of two capacitors, 4 μF and 6 μF in series:
- A) 10 μF
 - B) 2.4 μF
 - C) 1 μF
 - D) 5 μF
88. Combination of two capacitors, 4 μF and 6 μF in parallel:
- A) 10 μF
 - B) 2.4 μF
 - C) 1 μF
 - D) 5 μF
89. Energy stored in 2 μF capacitor at 10 V:
- A) 0.1 mJ
 - B) 0.5 mJ
 - C) 0.1 J
 - D) 1 J
90. Adding dielectric in capacitor:
- A) Increases stored energy at constant voltage
 - B) Reduces energy at constant voltage
 - C) Does not affect energy
 - D) Makes energy zero
91. Parallel plate capacitor has area 0.02 m², separation 1 mm, $\epsilon_0 = 8.85 \times 10^{-12}$ F/m. Capacitance is:
- A) 1.77×10^{-13} F
 - B) 1.77×10^{-10} F
 - C) 1.77×10^{-9} F
 - D) 1.77×10^{-12} F
92. Capacitance depends on:
- A) Charge
 - B) Geometry & dielectric
 - C) Voltage
 - D) Electric field
93. Two identical capacitors connected in series: total capacitance is:
- A) Half of one
 - B) Double of one
 - C) Equal to one
 - D) Four times one
94. Total charge on series combination is:
- A) Same on each
 - B) Sum of individual
 - C) Zero
 - D) Depends on voltage
95. Capacitance can be increased by:
- A) Increasing separation
 - B) Using dielectric
 - C) Reducing area
 - D) Removing dielectric

◆ SECTION 8: Gauss's Law & Electric Flux (Q96–120)

96. Gauss's law relates:
- A) Electric field & potential
 - B) Electric flux & enclosed charge
 - C) Force & charge
 - D) Current & voltage
97. Flux through closed surface enclosing charge Q :
- A) Q/ϵ_0
 - B) $Q \cdot \epsilon_0$
 - C) Zero
 - D) Depends on shape
98. Electric field inside conductor in electrostatic equilibrium:
- A) Zero
 - B) Maximum
 - C) Uniform
 - D) Non-uniform
99. Electric field due to infinite plane sheet:
- A) Varies with distance
 - B) Constant
 - C) Zero
 - D) Radial
100. Electric field due to infinite line charge:
- A) $1/r^2$
 - B) $1/r$
 - C) r
 - D) r^2
101. Gaussian surface should be:
- A) Any shape
 - B) Closed
 - C) Open
 - D) Flat
102. Flux through closed surface with no enclosed charge:
- A) Zero
 - B) Maximum
 - C) Minimum
 - D) Infinite
103. Flux is:
- A) Scalar
 - B) Vector
 - C) Tensor
 - D) Vector magnitude
104. Electric field outside uniformly charged sphere:
- A) Like point charge
 - B) Zero
 - C) Constant
 - D) Radial inside

105. Electric flux depends on:
A) Magnitude of field
B) Area
C) Angle between E & area
D) All above
106. Unit of electric flux:
A) $\text{N}\cdot\text{m}^2/\text{C}$
B) V/m
C) N/C
D) C/m^2
107. Which surface is best for Gauss's law calculation?
A) Any
B) Symmetric
C) Irregular
D) Non-closed
108. Flux through cube enclosing charge Q:
A) Q/ϵ_0
B) Zero
C) $Q/2\epsilon_0$
D) Depends on orientation
109. Field inside hollow conductor:
A) Zero
B) Maximum
C) Minimum
D) Uniform
110. Field inside solid conductor:
A) Zero
B) Varies linearly
C) Constant non-zero
D) Radial
111. Gauss's law is a consequence of:
A) Coulomb's law
B) Ohm's law
C) Newton's law
D) Faraday's law
112. Electric flux through plane perpendicular to field:
A) EA
B) $EA \cos\theta$
C) Zero
D) $EA \sin\theta$
113. Flux through plane parallel to field:
A) Maximum
B) Zero
C) Half
D) EA
114. Electric field due to thin charged shell inside:
A) Zero
B) Maximum

- C) Varies with r^2
 - D) Radial
115. Electric field due to thin charged shell outside:
- A) Like point charge
 - B) Zero
 - C) Uniform
 - D) Radial inside
116. For non-conducting solid sphere, field at center:
- A) Zero
 - B) Maximum
 - C) Half
 - D) Constant
117. Field inside spherical shell:
- A) Zero
 - B) Non-zero
 - C) Radial
 - D) Varies linearly
118. Field at surface of charged sphere:
- A) σ/ϵ_0
 - B) Zero
 - C) $E = kQ/r^2$
 - D) Both A & C
119. Electric flux is positive if field:
- A) Exits surface
 - B) Enters surface
 - C) Tangent
 - D) Perpendicular
120. Flux through cube with no enclosed charge:
- A) Zero
 - B) Non-zero
 - C) Maximum
 - D) Depends on orientation